Problem Set 5

Due Online: Friday, April 18, 11:59 pm

1. Consider the model for the $k \times 1$ vector $\mathbf{y}_t$:

$$\mathbf{y}_t = \mathbf{\Phi}_1 \mathbf{y}_{t-1} + \mathbf{\Phi}_2 \mathbf{y}_{t-2} + \boldsymbol{\varepsilon}_t, \tag{1}$$

where $\boldsymbol{\varepsilon}_t \sim WN(\mathbf{0}, \mathbf{\Sigma})$.

   (a) Write down the companion form of equation (1).

   (b) Using the companion form in part (a), derive the condition under which $\mathbf{y}_t$ is weakly stationary.
   (*Hint:* If $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are $n \times n$ matrices and $\mathbf{D}$ is nonsingular, then

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \det(\mathbf{D}) \det\left(\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}\right),$$

   where $\det(\cdot)$ denotes the determinant of a matrix.)

   (c) Find the coefficients matrices $\mathbf{\Psi}_1, \mathbf{\Psi}_2, \mathbf{\Psi}_3$ in the MA representation $\mathbf{y}_t = \sum_{i=0}^{\infty} \mathbf{\Psi}_i \boldsymbol{\varepsilon}_{t-i}$ as functions of the original matrices $\mathbf{\Phi}_1$ and $\mathbf{\Phi}_2$.

2. Consider the model

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} 0.4 & 0.6 \\ 0.25 & 0.5 \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} 0.1 & 0.4 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} y_{1,t-2} \\ y_{2,t-2} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix}, \tag{2}$$

where $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t})' \sim WN(\mathbf{0}, \mathbf{\Sigma})$.

   (a) Show that $\mathbf{y}_t = (y_{1t}, y_{2t})'$ is unit-root nonstationary. (*Hint*: Is one is a root of the determinant equation?)

   (b) Write equation (2) in the form of a vector error correction model (VECM).

   (c) Find the long-run relation between y_{1t} and y_{2t}, where the coefficient on y_{1t} is normalized to 1.

3. Consider the trivariate process:

$$\begin{pmatrix} x_t \\ y_t \\ z_t \end{pmatrix} = \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ -0.16 & 1.08 & 0.08 \\ 0.24 & -0.12 & 0.88 \end{pmatrix} \begin{pmatrix} x_{t-1} \\ y_{t-1} \\ z_{t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \\ \varepsilon_{3t} \end{pmatrix}, \tag{3}$$

where $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t}, \varepsilon_{3t})' \sim WN(\mathbf{0}, \boldsymbol{\Sigma})$, with $\boldsymbol{\Sigma}$ being a positive definite matrix.

(a) Write this model in the vector error correction form.

(b) How many cointegrating relations exist among $\{x_t\}$, $\{y_t\}$, and $\{z_t\}$? Find the cointegrating relation(s) and the speed of adjustment coefficients, normalizing the cointegrating matrix $\boldsymbol{\beta}$ such that its first m rows is the identity matrix, where m denotes the number of cointegrating relations.

(c) Given data on $\{x_t, y_t, z_t; t = 1, 2, \ldots, T\}$ and assuming standard regularity conditions hold, will OLS estimation of model (3) yield consistent and efficient estimates of the VAR coefficients? Explain why or why not.

4. (Computer Exercise) Consider the dynamic relationship between the quarterly real GDP and unemployment rate for the US. The levels of these two variables are contained in the file "quarterly.xls" under the names $RGDP$ and $Unemp$, respectively.

(a) Define the growth rate of real DGP as the logarithmic change $\Delta lgdp_t = \log(RGDP_t) - \log(RGDP_{t-1})$, and define the difference in the unemployment rate as $\Delta ur_t = Unemp_t - Unemp_{t-1}$. Determine whether $\Delta lgdp_t$ and Δur_t are stationary.

(b) Build a VAR model for $(\Delta lgdp_t, \Delta ur_t)'$ using the Akaike Information Criterion (AIC) to select the lag length. Write down the fitted model. Assess the adequacy of the model and explain your reasoning.

(c) Using the model obtained in part (b), compute the 1-step through 4-step ahead forecasts for $\Delta lgdp_t$ and Δur_t (assuming the forecast origin is the last data point).

(d) Do the two series $\Delta lgdp_t$ and Δur_t Granger cause each other?

(e) Plot the orthogonalized impulse response functions of the fitted model. Describe how each series responds to a shock in the other.

(f) Perform forecast error variance decomposition. What proportions of the forecast error variance in the unemployment rate are explained by GDP growth in the short run and long run?

5. (Computer Exercise) Retrieve monthly data for the following five U.S. Treasury interest rates from FRED using the `quantmod` package: 3-month (`TB3MS`), 6-month (`TB6MS`), 1-year (`GS1`), 5-year (`GS5`), and 10-year (`GS10`). Focus on the sample period from January 1981 to December 2024. Determine the number of cointegrating relationships among the five series using Johansen's trace and maximum eigenvalue tests. Report the estimated cointegrating vector(s).
Comment: For simplicity, assume all series are I(1) and skip formal unit root testing.